SIMATS ENGINEERING
ASSESSMENT TOOL 3: Class Test

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA05**

COURSE OUTCOME COVERED

CO2: ***Apply image enhancement and segmentation techniques for extracting meaningful regions from images.*** **(BTL 3)**

Assessment Tool Weightage: 10%

Code of Conduct:

I Harshavardhan. RV (192472163) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Harshavardhan

Signature of the student:

Rubrics for Evaluation

Criteria	*Weightage*	*Level 4 – Exemplary*	*Level 3 – Proficient*	*Level 2 – Developing*	*Level 1 – Beginning*	*Marks(100)*
Understanding of Concepts	*25%*	*Demonstrates thorough, accurate understanding of concepts*	*Good understanding with minor gaps*	*Basic understanding with some misconceptions*	*Poor understanding; major misconceptions*	
Explanation	*25%*	*Detailed, well-explained answers with relevant examples*	*Clear explanation with adequate details*	*Limited explanation; lacks depth*	*Poor or unclear explanation*	
Application	*20%*	*Effectively applies concepts with strong analytical ability*	*Applies concepts with minor errors*	*Limited application with weak analysis*	*No application or incorrect analysis*	
Organization	*15%*	*Well-structured answers with logical flow and coherence*	*Organized with minor issues in flow*	*Basic structure, lacks clarity*	*Poor organization, incoherent answers*	

Accuracy	*10%*	*Completely accurate and covers all required points*	*Mostly accurate with minor omissions*	*Partially correct with missing points*	*Incorrect answers with major gaps*	
Clarity	*5%*	*Clear, neat, professional presentation*	*Understandable with minor issues*	*Limited clarity, readability issues*	*Poorly written, difficult to understand*	

1.An image contains high-frequency noise.Explain and justify the appropriate frequency domain filter, including its working and impact on image quality.

ANS. Appropriate Filter: Low Pass Filter (LPF)

Explanation

- **High-frequency noise appears as rapid intensity changes in an image.**
- **A Low Pass Filter (LPF) removes these unwanted high-frequency components while preserving important low-frequency information.**

Working

1. **Convert the image to the frequency domain using the Fourier Transform.**
2. **Apply the Low Pass Filter.**
3. **The filter allows low frequencies to pass.**
4. **High-frequency noise is blocked.**
5. **Convert the filtered image back using the Inverse Fourier Transform.**

Impact on Image Quality

- **Reduces random noise.**
- **Produces a smoother image.**
- **Slightly blurs sharp edges.**
- **Improves overall visual quality.**

Justification

A Low Pass Filter is the best choice because it effectively removes high-frequency noise while retaining the important image content.

2.You need to preserve low-frequency components while removing noise. Describe and apply a suitable filtering technique, explaining how it achieves the desired effect.

Answer

Suitable Technique: Gaussian Low Pass Filter

Explanation

- **A Gaussian Low Pass Filter smoothly removes high-frequency noise.**
- **It preserves low-frequency information such as overall brightness and object shapes.**

Working

1. **Transform the image into the frequency domain.**
2. **Apply the Gaussian filter.**
3. **Frequencies farther from the center are gradually reduced.**
4. **Perform the Inverse Fourier Transform.**
5. **Obtain the filtered image.**

Advantages

- **Smooth noise reduction.**
- **Less ringing effect.**
- **Preserves image appearance.**
- **Produces natural smoothing.**

Conclusion

The Gaussian Low Pass Filter removes unwanted noise while maintaining the essential low-frequency components.

3.An image requires edge enhancement in the frequency domain. Explain and justify the filtering approach used, describing its effect on frequency components.

Answer

Appropriate Filter: High Pass Filter (HPF)

Explanation

- **Edges contain high-frequency information.**
- **A High Pass Filter enhances these high-frequency components.**

Working

1. **Convert the image into the frequency domain.**
2. **Apply the High Pass Filter.**
3. **Low frequencies are suppressed.**
4. **High frequencies are preserved.**
5. **Convert back using the Inverse Fourier Transform.**

Effect

- **Sharpens image details.**
- **Enhances edges.**
- **Improves object boundaries.**
- **May increase image noise.**

Justification

High Pass Filtering is suitable because it highlights fine details and edge information effectively.

4.A smooth version of an image is required.Describe and justify the frequency domain technique used for smoothing, explaining its operation and results.

Answer

Technique: Ideal Low Pass Filter

Explanation

- **Smoothing removes unwanted fine details and noise.**
- **An Ideal Low Pass Filter passes only low-frequency components.**

Working

1. **Perform Fourier Transform.**
2. **Apply the Ideal Low Pass Filter.**
3. **Remove high-frequency components.**
4. **Perform Inverse Fourier Transform.**
5. **Obtain a smooth image.**

Results

- **Noise is reduced.**
- **Image becomes smoother.**
- **Edges become softer.**
- **Fine details decrease.**

Justification

The Ideal Low Pass Filter effectively smooths the image by removing high-frequency information.

5.An object is clearly brighter than the background. Explain and justify the segmentation technique used, including its working and effectiveness.

Answer

Technique: Threshold Segmentation

Explanation

- **Threshold segmentation separates objects based on intensity.**
- **Pixels above a threshold belong to the object.**
- **Pixels below belong to the background.**

Working

1. **Select a threshold value.**
2. **Compare each pixel intensity.**
3. **Bright pixels become foreground.**
4. **Dark pixels become background.**
5. **Generate the segmented image.**

Effectiveness

- **Fast and simple.**
- **Accurate for high contrast images.**
- **Easy to implement.**

Justification

Threshold segmentation is ideal because the object is significantly brighter than the background.

6.You need to separate foreground and background based on intensity. Describe and apply an appropriate segmentation method, providing a clear explanation of its process.

Answer

Method: Global Thresholding

Explanation

- **Global Thresholding uses one threshold value for the entire image.**
- **It separates foreground and background based on pixel intensity.**

Process

1. **Choose a threshold value.**
2. **Compare every pixel with the threshold.**
3. **Pixels above threshold become foreground.**
4. **Remaining pixels become background.**
5. **Produce a binary image.**

Advantages

- **Simple implementation.**
- **Fast computation.**
- **Effective when illumination is uniform.**

Conclusion

Global Thresholding accurately separates foreground and background in images with clear intensity differences.

7.An image requires identification of object edges. Explain and justify a suitable edge detection method, describing its working and effectiveness.

Answer

Method: Canny Edge Detection

Explanation

- **Canny is one of the most accurate edge detection techniques.**
- **It detects strong and continuous edges.**

Working

1. **Smooth the image using a Gaussian filter.**
2. **Calculate intensity gradients.**
3. **Apply Non-Maximum Suppression.**
4. **Use Double Thresholding.**
5. **Track edges using hysteresis.**

Effectiveness

- **Detects accurate edges.**
- **Reduces noise.**
- **Produces thin and continuous edges.**
- **Minimizes false detections.**

Justification

Canny Edge Detection provides reliable and precise edge detection for most images.

8.Adjacent pixels have similar intensity values. Describe and apply a segmentation technique to group such pixels, explaining the concept and reasoning.

Answer

Technique: Region Growing

Explanation

- **Region Growing groups neighboring pixels with similar intensity.**
- **It starts from selected seed points.**

Working

1. **Select a seed pixel.**
2. **Check neighboring pixels.**
3. **Add similar pixels to the region.**
4. **Repeat until no similar neighbors remain.**
5. **Form segmented regions.**

Advantages

- **Produces connected regions.**
- **Preserves object shape.**
- **Accurate for homogeneous areas.**

Justification

Region Growing effectively groups pixels having similar intensity into meaningful regions.

9.Detected edges in an image appear broken.
Explain and justify a method to improve edge continuity, including its role in post-processing.

Answer

Method: Morphological Closing

Explanation

- **Broken edges reduce segmentation accuracy.**
- **Morphological Closing connects nearby edge gaps.**

Working

1. **Perform Dilation.**
2. **Expand edge boundaries.**
3. **Perform Erosion.**
4. **Restore original edge thickness.**
5. **Obtain connected edges.**

Role in Post-processing

- **Connects broken edges.**
- **Removes small gaps.**
- **Improves boundary continuity.**
- **Enhances segmentation accuracy.**

Justification

Morphological Closing is effective because it connects discontinuous edges while preserving object shape.

10.You need to extract object outlines from an image. Describe and apply an appropriate technique, explaining how it identifies boundaries effectively.

Answer

Technique: Boundary Extraction using Morphological Operations

Explanation

- **Boundary Extraction identifies object outlines.**
- **It uses erosion followed by subtraction.**

Working

1. **Obtain the binary image.**
2. **Apply Morphological Erosion.**
3. **Subtract the eroded image from the original image.**
4. **Remaining pixels represent object boundaries.**
5. **Display the extracted outline.**

Advantages

- **Produces clear boundaries.**
- **Preserves object shape.**
- **Simple implementation.**
- **Useful for object analysis.**

Justification

Morphological Boundary Extraction accurately identifies object outlines while preserving the original object structure.